

A Robust Video Based License Plate Recognition System

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Abstract— License plate recognition system (LPRS) is a difficult problem in the field of machine vision and is of substantial interest because of its applications to areas such as cross border security, law enforcement and various other automation applications. Previous methods used plate specific details such as aspect ratio, color or dimensions of the plate in the complex task of plate localization. In this paper, license plate localization process is carried out by weight based density map (WBDM) method, which is independent of such constraints. The proposed method also relaxes constraints in lighting conditions. The robustness of this method is well suited for application in countries like India where the appearance of plates varies widely. 1750 images were captured in different background scenes under various lighting conditions. This method has been tested with different types of vehicles such as buses, cars, trucks etc. The overall identification rate of this algorithm is 98.63% in different conditions.

Index Terms— Image analysis, edge detection, character segmentation, character recognition, license plate localization.

I. INTRODUCTION

LICENSE plate recognition performs an imperative responsibility in several transportation applications. Recognizing the license plate of a vehicle from a natural image is a complicated process that involves the detection of the license plate and recognition of characters on the plate. LPRS techniques differ in accordance with different types of applications. Many techniques have already been suggested as in [1], [2]. However, these systems are not viable in developing nations like India. Most of these techniques put a lot of constraints on the size and color of the plate, the size of the characters, position of the vehicle, stationary backgrounds and fixed illumination. The primary objective of this work is to minimize the number of restrictions and produce a generalized solution for LPRS that is aptly suited for Indian working condition. The proposed system consists of four phases whose functions are briefly described as follows:

- Image acquisition and Quality Frame Selection: A video camera attached to a frame grabber gives an image sequence with a resolution of 640x480. From this sequence, the frame with best quality is taken.
- Pre-processing: This phase consists of color subtraction and edge detection, which is used to

eliminate unwanted data.

- License plate localization: This phase extracts license plate candidate regions from a complex scene.
- Character Segmentation and recognition: Segmentation is carried out using blob-coloring method supported with constraint checking. The segmented characters are then scaled and fed into the neural network for recognition.

The organization of this paper is as follows: Section II summarizes the literature on the previous research. The proposed method for License plate recognition is illustrated Section III. Section IV illustrates License number recognition process. Experiment and Results are given in Section V. Section VI gives the concluding remarks on the work.

II. REVIEW OF LITERATURE

License plates (LP) have been localized by looking for rectangular LP edges in the image edge map. The prerequisite for this method is that the license plate should exhibit sharp rectangular edges. However, if the color of the car and the license plate are almost the same and the transition from car body to the LP is not very distinguishable, in that case the rectangular edges of the license plate will not be visible. Therefore LP localization using only the edge map is not very robust. In [3] color histograms were used to locate the license plate. This technique fails under different lighting conditions when the actual color of the LP differs from the color reflected by it. This problem was overcome by the use of a neural network to first classify the color. Still the system would fail when the color of the license plate is the same as other parts of the car, or produces a similar histogram. In [4], [5], [6], [7], [10] plate specific parameters such as plate aspect ratio or even exact plate measurements are used to locate correct candidate region. This method would fail when the aspect ratios vary drastically as observed in Indian license plates. Systems like [2] utilize intelligence like neural networks to process the image and localize the license plate. Other methods as in [1] that have been aimed at relaxing the constraints on the complexity of the images have been successful in achieving lesser false detection rates. But the system remains to be improved further. Thus many effective techniques have been proposed and claim to be able to work with other kinds of license plates without making any major

design changes. But as elaborated above, Indian license plate recognition requires an approach that is suited to a wide variation of plate parameters such as color, font size and style, plate dimensions etc.

III. LPRS ALGORITHM

The block diagram of the LPRS is shown in Fig 1. The vehicles' video sequence (25 frames/s) acquired by the camera is sent to the LPRS. The input sequence is first digitized into individual frames that have a resolution of 640 X 480 pixels. Quality Frame Selection (QFS) module selects an appropriate frame from the input video sequence and passes it to the color subtraction module. An LPRS requires only image data contained in the actual license plate. The image pre-processing module removes unnecessary data and thus saves a large amount of time. Edge detection module distinguishes all possible edges in the image. The license plate region is characterized by a dense collection of edges formed due to the difference in intensity of the dark foreground and light background of the plate. WBDM highlights these regions of high edge density.

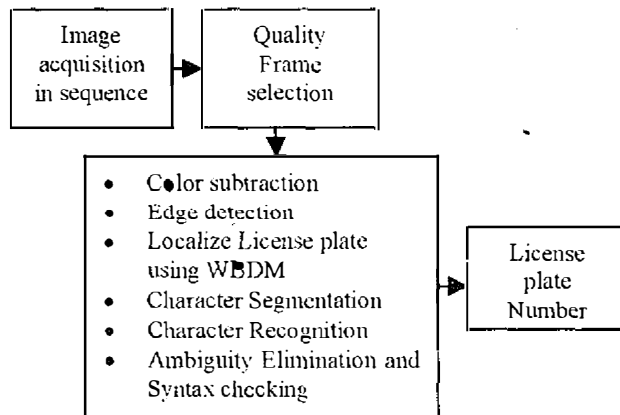


Fig.1 Block diagram of the LPRS

There are two important tasks involved in the license plate identification process, character segmentation and recognition. The final module verifies the characters by means of syntax checking procedures and produce accurate license number. Each of the above mentioned processes will now be discussed in detail.

A. Image acquisition and Quality frame selection

This module is responsible for acquiring image sequence from a digital camera. The sequences are captured using a video camera with a 640x480 pixel resolution. License plates may vary in color, size, position, and orientation. The image acquisition system outputs a sequence of the captured frames. For applications involving LPR of vehicles at high speeds, a camera with a very high shutter speed is required in order to prevent motion blurring. Such cameras would greatly increase the cost of the system. However, in applications such as secure access control and automated tollgates, we may opt for an ordinary still camera. The entire process would be much

faster if LPR could be done when vehicle is slowly on the move, thereby reducing traffic delays. We propose a motion camera with a frame grabber for such applications. This frame selection module picks the best frame from the input sequence and then proceeds. This method proves to be more robust than a still camera capturing just a single frame. In the input sequence, not all vehicle images are sharp enough for license plate recognition. We present an effective scheme to appraise image feature by analyzing the frequency distribution of the vehicle image. This method is capable of differentiating between clear images and blurred images. Fig. 2 shows variation of low and high quality images and their Fourier spectra.

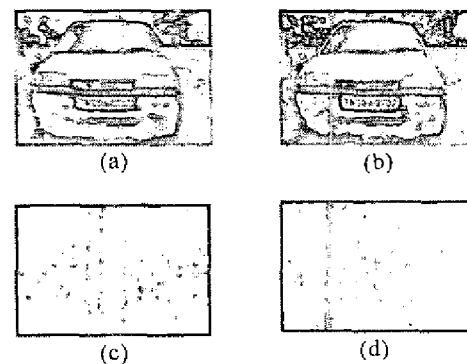


Fig.2 Variations of low quality and high quality images.
(a) Low quality image (b) high quality image (c) and (d) Fourier spectrum of (a) and (b) respectively

The block diagram of QFS is shown in Fig.3. The feature descriptor (FD) consists of two discriminate frequency features. The first feature is the total spectrum power of a license plate region, which can effectively discriminate clear license plate images from defocused vehicle images. The second feature is the ratio of the middle frequency power to other frequency power. It should be larger for clear images than motion blurred images. Eq.1-3 describes the QFS process.

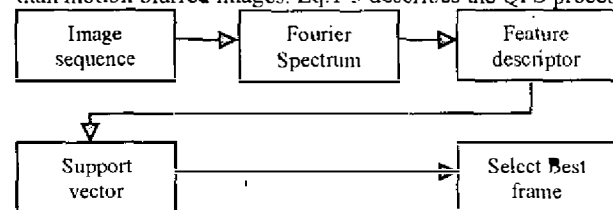


Fig.3 Block diagram of Quality Frame selection module

$$F_i = \int_{-\infty}^{\infty} \int_{-\infty}^{\infty} |F(u,v)| du dv \quad i=1,2,3,\dots \quad (1)$$

$$|F(u,v)| = f(x,y) XY |\sin(\pi u X) / \pi u X| |\sin(\pi v Y) / \pi v Y| \quad (2)$$

$$FD = (F_1 + F_2 + F_3, F_2 / (F_1 + F_3)) \quad (3)$$

where, $|F(u,v)|$ is the 2D Fourier spectrum of an vehicle image, F_1 , F_2 , and F_3 are the power of low, middle and high

frequencies respectively. Support vector machine is used to distinguish clear and motion blurred vehicle images. This method separates data from two categories by optimal hyper plane. Here we assume each pattern x_k has been transformed to $y_k = \phi(x_k)$, where $\phi(\cdot)$ is an appropriate nonlinear mapping. For each of the n patterns $k=1, 2, 3, \dots, n$, let $z_k = +1$ or -1 , according to pattern k in class ω_1 or class ω_2 . In this case, ω_1 and ω_2 are called features of clear and motion blurred images respectively. Clear image forms a compact cluster in the feature space.

B. Image Pre-Processing

The only relevant image data required for license plate recognition is the license plate of the subject vehicle. Eliminating unnecessary data greatly save time and boosts the efficiency of the system. This task is performed by color subtraction followed by edge detection. The image is processed in a bottom-up fashion, as the license plate tends to be in the lower areas of the image. Thus input image is flipped vertically before further processing.

1) Color Subtraction

The license plates are of two possible combinations namely black foreground with white background and black foreground with yellow background. Color subtraction retains all pixels having these color values for each color and blackens out the rest of the image. A range of RGB values was used for each color instead of absolute values. This compensates for the change in shades of the colors caused by different lighting conditions. For example, at dusk white appears to have a slight red tinge. The remaining image contains only essential data for the further phases of license plate recognition. The Eq.4 - 5, summarizes this process.

$$f(x,y) = a_{x,y} \quad (4)$$

$$g(a_{x,y}) = \begin{cases} 0 & \text{if } a_{x,y} \leq T \\ S & \text{if } a_{x,y} > T \end{cases} \quad (5)$$

Then, $(g \circ f)$ is a composition function of f and $g = h$, from (x, y) to $(0, S)$ where f is a function from ordered pair to pixels in the image, g is a function from pixels to 0 or S . $a_{x,y}$ are a set of pixel values in the image. T is a threshold value of R, G and B values of each pixel. S is the unchanged RGB value and h is a composition of function.

2) Edge Detection

The LP localization process relies on edge information of the LP characters. Edge detection can be carried out by convolving the image with kernel masks like the Sobel filter mask. We propose an edge mapping process using the threshold method as in Eq.6-7. Typical output color subtraction and edge detection is shown in Fig.4.

$$g(x,y) = a_{x,y} \quad (6)$$

$$e(a_{x,y}) = \begin{cases} 0 & \text{if } 0 \leq |(a_{x-1,y} - a_{x,y})| \leq \text{LIT} \\ 255 & \text{if } \text{LIT} < |(a_{x-1,y} - a_{x,y})| \leq \text{HIT} \end{cases} \quad (7)$$

Then, $(e \circ g)$ is a composition function of g and $e = h$, from (x,y) to $(0, 255)$ where g is a function from ordered pair to pixels in the image, e is a function from pixels to 0 or 255. $a_{x,y}$ are a set of pixel values in the color subtracted image. LIT and HIT are lower and higher intensity thresholds respectively. Applications with external illumination have constant LIT and HIT values. These values may change in application where lighting conditions vary drastically. This enables the system to compensate for varied lighting conditions i.e. time of day.

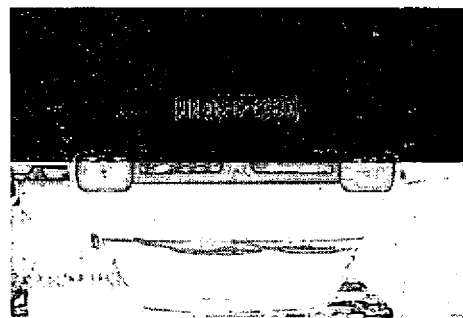


Fig.4 Sample output after invoking color subtraction and edge detection process.

C. License Plate Localization

This process is critical to the success of any LP recognition system. Many techniques have been proposed for this task. Some techniques use the color of plate to detect it [3]. Most of the plate localization algorithm rely on plate specific properties such as looking for rectangles with standard aspect ratios [4],[5],[6],[10],[11] of license plates or looking for edges that are equal to the standardized length or width of license plates[7]. These techniques cannot be adopted in countries where strict regulations on plate parameters have not been imposed. For example, Indian license plates exhibit a wide variation in their aspect ratios as shown in Fig 5.

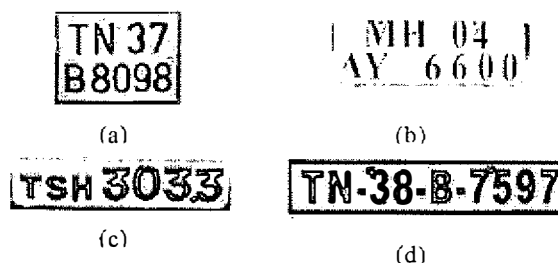


Fig.5 Typical license plate aspect ratio variation
(a) 1.3:1 (b) 3:1 (c) 4:1 (d) 6:1

The proposed method for LP localization involves the following. First, the Weight Based Density map method locates regions of high edge density and the coordinates of these rectangular regions are stored. Next, various criteria are checked to filter out the rectangular regions that are possible license plates.

1) Weight Based Density Map

This method exploits the fact that, the edges are denser in the region containing the license plate than any other region of

the image. To determine the concentration of edges the weight assignment scheme described in Eq. 8 – 12 is applied. The weight allocation is done in a manner such that the degree by which the weight increases or decreases is directly proportional to the distance between current edge and next edge. Similarly the decrease in weight is directly proportional to the step size of each weight increment. This equation is derived as follows:

It can be stated that Weight decrement is directly proportional to $(i - p)$. Therefore,

$$W_{t_decrement} \propto (i - p) \quad (8)$$

Weight decrement is directly proportional to γ . Therefore,

$$W_{t_decrement} \propto \gamma \quad (9)$$

From (8) and (9), we can derive (10)

$$W_{t_decrement} = \partial \gamma (p - i) \quad (10)$$

Where ∂ is decay constant. Therefore,

$$W_t = \sum_{i=0}^{n-1} t_w (W_t + (px_i)\gamma + (1-px_i)(\partial \gamma (p-i))) \quad (11)$$

$$t_w = (((W_t - |W_t|) / 2W_t) \beta) + (W_t + |W_t|) / 2W_t \quad (12)$$

Where, n is number of coordinate points, i is a coordinate of the current edge, p is a coordinate of the previous edge, γ is a Weight increment factor, β is an initial weight and $W_{t_decrement}$ is a weight decrement factor and W_t is a weight value.

WBDM is expressed in an algorithmic notation below

Step1: Assign current pixel position $i = 1$, $w[i] = 0$, $wt = 0$

Step2: if current pixel is WHITE and $wt < 1$ then $t = i$ and $wt = \beta$ otherwise $p = i$, $wt = wt + \gamma$, $w[i] = wt$

Step3: if current pixel is BLACK and $wt > 1$ then $wt = wt - \partial \gamma (p - i)$, $w[i] = wt$

Step4: Repeat step2 and step3 until $i \geq N$

W_t represents the continuously varying weight of each row of the edge map and depends on the density of edges in the area under scrutiny. A raster scan begins at the beginning of every row of the edge map. When the first edge pixel is found, an initial weight of β is assigned to W_t . This weight is then associated with the particular pixel (as $W[n]$) and next pixel is investigated. If it is an edge pixel, the weight W_t is incremented by the weight increment factor γ and then it is allocated to the current pixel as $W[n]$. Also location of this pixel is stored in p for use in the calculation of weight decay. If however the next pixel is black, then W_t is decremented as shown in Step3. Notice, that the degree by which the weight W_t decays depends on both the decay constant ∂ as well as the distance between current pixel and most recently encountered edge. ' p ' holds the location of most recent pixel and ' i ' holds the location of current pixel. As a result, the weight decays at a much faster rate where edges are sparse as compared to regions of dense edges. This weight is then allocated to the current pixel as $W[n]$. The procedure is repeated until all pixels in the current row have been exhausted. The process is repeated for all remaining rows of the edge map. Fig.6 shows the weight variation for a particular row of the edge map. The peaks denote areas of concentrated edges. The areas of high edge density obtained in the WBDM were compared to the corresponding areas in the input images and we arrived at a threshold value that can be used to extract the possible plate

regions. As the algorithm progresses, lines are drawn over areas of concentrated edges. These lines are simply meant to act as markers. The implementation of this scheme produced best results for the values $\partial = 0.015$, $\gamma = 2$ and $\beta = 6$.

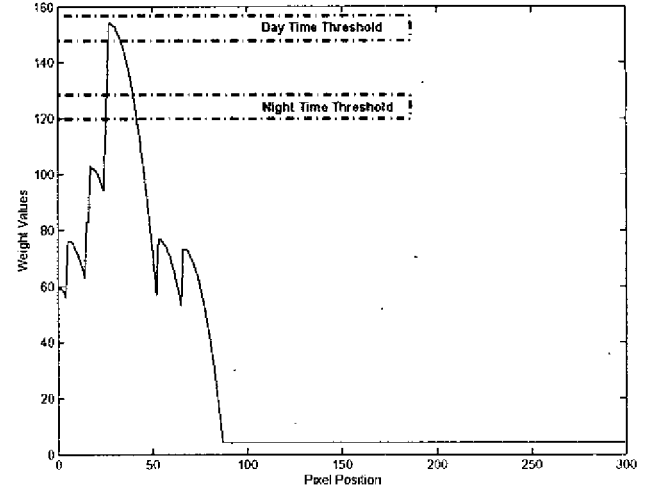


Fig.6 Graph showing typical weight variation

2) Deriving Plate Coordinates

Once the weight based density map algorithm has scanned the lower half of the image, we are left with a view of the image that highlights areas of high edge concentration. The lines of approximately equal length sharing approximately the same horizontal components but vary in their vertical component are first grouped together and their extremity values (top left and bottom right coordinates) of each collective group of lines are stored. The result is a sequence of rectangular regions that represent possible license plate regions. In order to avoid clipping of characters on the plate, we expand the rectangles until there are at least two lines of white pixels on all four sides.

IV. LICENSE NUMBER RECOGNITION

A. Character Segmentation and Extraction

A popular scheme for segmentation is the histogram method [3], [8] in which the characters are segmented in the region between two crests of the histogram. However, this scheme would fail if the symbols on the license plate appear in two lines as shown in Fig.5a. Thus the proposed approach uses Blob-coloring algorithm and is supplemented with constraint checking. During Blob-coloring, the size in terms of pixels and also the extremity coordinates of each blob is recorded in a blob descriptor table (BDT). The segmentation routine continues iteratively until all blobs within given rectangular region have been investigated as shown in Fig.7. This process is repeated for all possible rectangular license plate regions extracted in the previous phase.

In order to zero in on the rectangle that corresponds to actual plate, we check whether the blobs have dimensions (found in BDT) similar to that of license plate characters. This

acceptable range was derived from a field study of the various possible character sizes. This constraint checking process eliminates the chance of blobs occurring due to dust, black spots and other designs on the license plate to be classified as license plate characters.

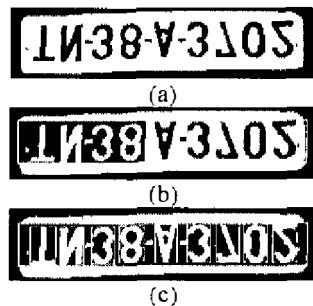


Fig.7. Character Segmentation
(a) Binarized plate region. (b) Blob Coloring in process.
(c) Segmentation of characters based on Blob properties.

B. Character Recognition

Typically template matching is the method of choice in most license plate recognition as in [3], [4], [9], [12] Template matching was experimented with initially, but due to wide variation of character font, the results were not satisfactory. Finally a neural network based character classifier was chosen. The segmented symbols are first binarized, scaled to a size of 14x10, and fed into the neural network for character recognition. We use a conventional BPN as the recognition engine. The network has 140 inputs that it acquires from the 14x10 segmented symbols and has 36 outputs.

C. Ambiguity Elimination Module:

We observed that certain ambiguous characters were wrongly classified, for example pairs like 'H' and 'N'. Whenever such problematic characters are encountered, the minute distinguishing features are compared to differentiate between the ambiguous pair. An example of these minute features has been shown in the Fig.8.



Fig.8. Ambiguity elimination between 'H' and 'N'

Even at this stage there may be confusion as to which rectangle is the actual license plate. This means that without further verification, a rectangular area having a sticker may be mistaken for the license plate. Thus after recognition is completed, we go for a final verification process that involves syntax analysis. We check to see if the string of characters represents a meaningful license plate number. At the end of this process we successfully zero in on one particular rectangular region, which we can confidently say is the license plate of the car.

V. EXPERIMENTAL RESULTS

The implementation has been done in VC++ and the time efficiency has been shown based on observations made on an Intel Pentium IV 2.4 GHz machine with 256 MB RAM. The efficiency of the proposed system has been tested with a wide variety of input images. Our image database contains 1750 vehicle images sequences with resolution 640x480 pixels. Example image database is shown in Fig.9.

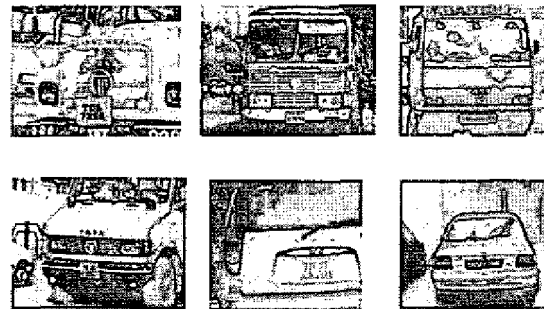


Fig.9 Sample Images in the database

The images were taken from various environments under different illumination. We captured images at various distances (D1-D7). Distance was varied from 2 feet to 20 feet in steps of 3 feet. The camera was maintained at a height of 2 feet above ground level.

TABLE 1: LICENSE PLATE LOCALIZATION RESULTS

Vehicle and No. of Images	Success Rate At Various Distances (in %)							Avg. Success per Type
	D1	D2	D3	D4	D5	D6	D7	
Car 1200	100	100	99.8	99.6	98.4	98.6	98.3	99.2
Truck 100	100	99	100	98	98	98	97	98.6
Bus 250	100	100	99.6	99.6	97.6	96.8	96.8	98.6
Van 200	99.5	99.5	99.5	99	98.5	96	95.5	98.2
Avg. Rate%	99.8	99.6	99.7	99.1	98.1	97.3	96.9	98.65

We collected a sample database of 1750 images for testing the system to effectively measure the accuracy of the system at various distances since distance variation could not be accurately calculated from real time video sequences. The system was finally tested with a number of image sequences. The system is robust for angle variations between 40 and 90 degrees. Angles below 40 degrees cause problems in segmentation, as the symbols seem joined. Quality frame selection is based on support vector machine. A compact cluster in the feature space is used as an indication of a high quality image as shown in Fig.10.

The clarity of symbols in the license plate determines the accuracy of the segmentation module. If the symbols have space of at least one pixel width in between, they can be segmented efficiently. However, this is not always the case in India.

In some plates, the symbols are smudged together or other foreign materials may partially occlude the plate as shown in Fig.11. The efficiency of the character recognition remains constant since it is independent of the distance as shown in Table 2.

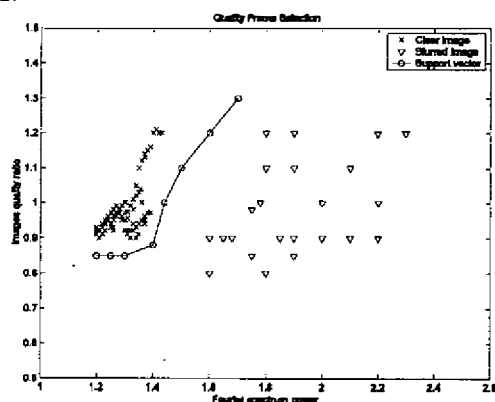


Fig.10 Quality frame analysis in feature space



Fig.11 Example of partially occluded license plates

The error in license plate localization is 1.35 %. The error in character recognition is 2.05 %. When these two are superimposed an overall error rate of 3.37 % is obtained. Hence the overall success rate is 96.63 %. Table 3 illustrates time complexities of various processes performed in the proposed system.

TABLE 2: CHARACTER RECOGNITION RATE

Type of Vehicle	Character Recognition Rate
Car	98.3 %
Truck	97.7 %
Bus	97.6 %
Van	98.2 %
Average Rate	97.95 %

TABLE 3: TIME COMPLEXITY

Critical Processes in LPRS	Time (seconds)
Frame Selection	0.9
Image Preprocessing	1.2
LP Localization	0.9
Character Segmentation	0.7
Character Recognition	2.2
Total	5.9

V. CONCLUSION

In this paper we have proposed and examined a new methodology for license plate recognition. The proposed methodology succeeds in relaxing a number of constraints like the distance between the vehicle and camera, the color of the license plate, the fonts of the license plate characters etc. The

recognition rates in various conditions have been studied and have been illustrated in the paper. The proposed methodology has been found to show high recognition rates and quite suitable for conditions where a wide variety of license plates are to be dealt with.

Future work in this area can concentrate on extending this methodology to extract and recognize multiple plates from the same image. This could be done by using adaptive threshold for the weights during localization and segmentation process.

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